

HCIC Non-Python Study Guide

Tier 1 Topics • Appears Every Year

This guide covers the four non-Python topics that have appeared in every HCIC paper from 2021 to 2024. Master these first; they account for the majority of non-Python marks.

The Four Tier 1 Topics

1. Gear RPM Calculations
2. Robot Stability and Centre of Gravity
3. Line Following Logic
4. Flowcharts

Topic 1 — Gear RPM Calculations

Gears are used to transfer rotational motion from one shaft to another. The key relationship is between the number of teeth on each gear and how fast it spins.

The Core Formula

Gear Formula

$$\text{Driver Teeth} \times \text{Driver RPM} = \text{Driven Teeth} \times \text{Driven RPM}$$

$$\text{Rearranged: } \text{Driven RPM} = (\text{Driver Teeth} / \text{Driven Teeth}) \times \text{Driver RPM}$$

The Intuition

- A SMALLER driven gear spins FASTER than the driver (more RPM, less torque).
- A LARGER driven gear spins SLOWER than the driver (less RPM, more torque).
- Think of it like a bicycle: small front chainring + large rear sprocket = easier pedalling (more torque), but you spin the wheels slower.

Speed vs Torque Trade-off

Condition	Effect on Driven Gear
Driven gear is SMALLER	Higher RPM, lower torque (speed up)
Driven gear is LARGER	Lower RPM, higher torque (slow down)
Same size gears	Same RPM, same torque

Worked Example — Two Gears

Gear A has 30 teeth, Gear B has 20 teeth. Gear A makes 1 revolution. How many does B make?

Step-by-step

Formula: $\text{Driver Teeth} \times \text{Driver RPM} = \text{Driven Teeth} \times \text{Driven RPM}$

Plug in: $30 \times 1 = 20 \times \text{Driven RPM}$

Solve: $\text{Driven RPM} = 30 / 20 = 1.5 \text{ revolutions}$

Direction: B is smaller $\rightarrow$ B spins faster. Answer: 1.5 revolutions, B spins faster.

Worked Example — Chain of Three Gears

Gear A: 20T at 60 RPM $\rightarrow$ Gear B: 60T $\rightarrow$ Gear C: 10T. Find Gear C's RPM.

Step-by-step

Step 1 — A to B: $20 \times 60 = 60 \times \text{B_RPM} \rightarrow \text{B_RPM} = 1200 / 60 = 20 \text{ RPM}$

Step 2 — B to C: $60 \times 20 = 10 \times \text{C_RPM} \rightarrow \text{C_RPM} = 1200 / 10 = 120 \text{ RPM}$

Answer: Gear C spins at 120 RPM.

Key rule: solve one pair at a time, left to right through the chain.

Worked Example — Hub Gears

Drive Gear: 40T at 1 RPM. Find RPM for 24T, 20T, and 8T gears all meshing with the 40T gear.

Step-by-step

24T gear: $40 \times 1 = 24 \times \text{RPM} \rightarrow \text{RPM} = 40/24 = 1.67 \text{ RPM}$

20T gear: $40 \times 1 = 20 \times \text{RPM} \rightarrow \text{RPM} = 40/20 = 2 \text{ RPM}$

8T gear: $40 \times 1 = 8 \times \text{RPM} \rightarrow \text{RPM} = 40/8 = 5 \text{ RPM}$

Answer matches option (C): 1.66, 2, 5.

Quick Reference Table

Driver Teeth	Driven Teeth	Effect
30T	20T	Driven spins 1.5× faster
20T	60T	Driven spins 3× slower
40T	8T	Driven spins 5× faster
15T	15T	Same speed

Common Exam Traps

- **Chain of gears:** solve one pair at a time. The middle gear acts as both driven (from the left) and driver (for the right).
- **Three-gear loop:** if gear A drives B which drives C which drives A, the gears are locked and cannot move
- **Large gear = more torque:** for load-hauling or hill-climbing robots, you want a larger driven gear (slower but stronger). For speed, a smaller driven gear.

Gear Rotation Direction

When gears mesh, they push against each other. The direction of rotation alternates at every mesh point. This topic tests whether you can trace rotation direction through a gear train.

The One Rule You Need

Direction Rule

When two gears MESH, they rotate in OPPOSITE directions.

In a chain: $A \rightarrow B \rightarrow C \rightarrow D \dots$

A and C rotate the same direction (two meshes cancel out).

A and B rotate in opposite directions (one mesh).

The direction alternates at every gear in a simple chain.

Odd and Even Position Rule

In a simple gear chain, gears in ODD positions (1st, 3rd, 5th...) rotate in the same direction as each other. Gears in EVEN positions (2nd, 4th...) rotate in the same direction as each other, but opposite to the odd-position gears.

Gear	Position	Direction (if Gear 1 = Clockwise)
A (Driver)	1st	Clockwise
B	2nd	Counter-clockwise
C	3rd	Clockwise (same as A)
D	4th	Counter-clockwise (same as B)

The Gear Loop Trap

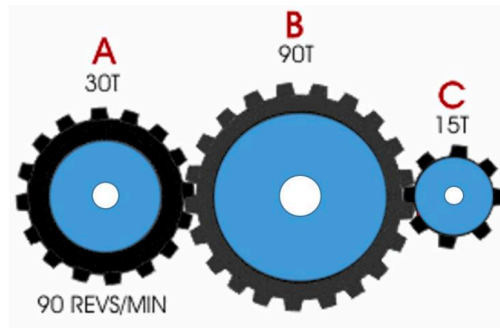
Locked Gear System

If gears form a CLOSED LOOP (A meshes B meshes C meshes back to A), the system is LOCKED and CANNOT MOVE.

Why: Gear A tries to push B clockwise, B tries to push C counter-clockwise, but C then tries to push A counter-clockwise, contradicting A's own motion.

Worked Example

The figure below shows a gear train of THREE GEARS, which of the following statements is true?



- (A) The gear ratio of A : B is 2 : 1
- (B) Gear B and C has more torque than speed.
- (C) If the input Gear A moves clockwise, the output Gear C will move counter-clockwise.
- (D) If Gear B moves counter-clockwise, Gear A and Gear C will move clockwise

Checking each option

(A) Gear ratio A:B is 2:1 — FALSE. A has 30T, B has 90T. Ratio is $30:90 = 1:3$.

(B) Gear B and C have more torque than speed — FALSE. C is smaller than B, so C has more speed.

(C) If A moves clockwise, C moves counter-clockwise — FALSE. A→B is opposite, B→C is opposite again.

So C rotates the SAME direction as A (clockwise). Two opposites cancel.

(D) If B moves counter-clockwise, A and C both move clockwise — TRUE.

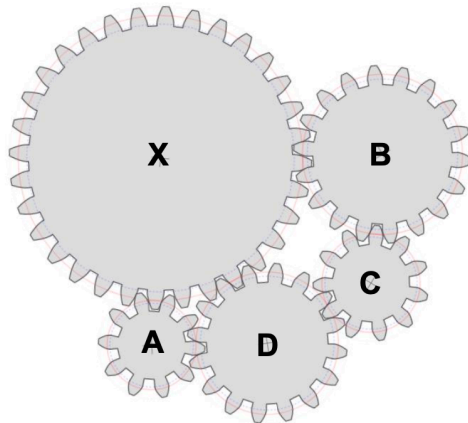
B counter-clockwise → A clockwise (opposite). B counter-clockwise → C clockwise (opposite).

Answer: (D)

Worked Example

Four gears X, B, A, D are arranged so that X meshes B, B meshes with both C and A, C meshes D, and D meshes back toward A. Which statement is true?

Observe the following gear structure carefully. Which of the following statements is true?



- (A) The structure is moving smoothly with all gears in action.
- (B) The structure should move only if gear B is removed.
- (C) The structure should move only if gear C is removed.
- (D) The structure is unable to move.

Answer

The gear arrangement forms a loop — B drives C, C drives D, and D connects back.

This creates a contradiction in rotation directions.

Answer: (D) The structure is unable to move.

Topic 2 — Robot Stability and Centre of Gravity

The Centre of Gravity (CG) is the single point where all of an object's weight acts. Its position determines how stable the object is.

The Two Rules of Stability

Stability Rules

Rule 1: LOWER centre of gravity = MORE stable.

Rule 2: WIDER base = MORE stable.

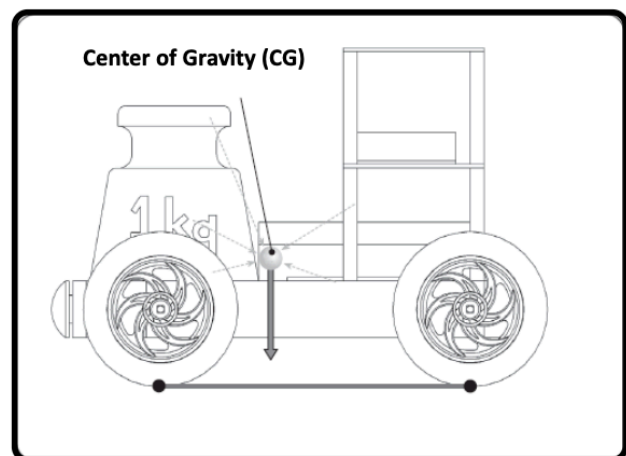
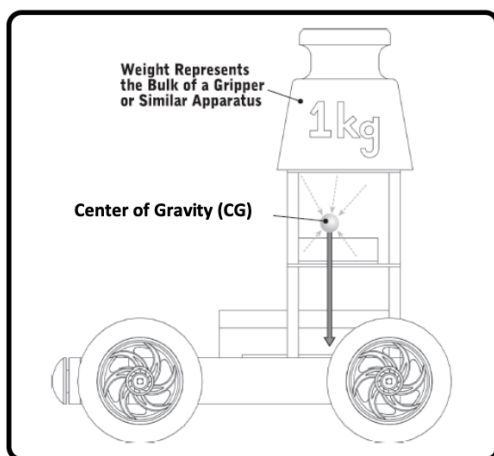
A robot tips over when its CG moves outside the boundary of its base.

Effect of Weight Placement

Weight position	Effect on stability
Weight mounted HIGH (e.g. top of robot)	CG rises → LESS stable → easier to tip
Weight mounted LOW (e.g. base of robot)	CG drops → MORE stable → harder to tip
Weight spread wide	Wider effective base → MORE stable
Weight concentrated narrow	Narrower base → LESS stable

Worked Example

Bot A has a 1kg weight mounted high (CG shown high). Bot B has the same weight mounted low (CG shown low). Which is more stable?



Answer

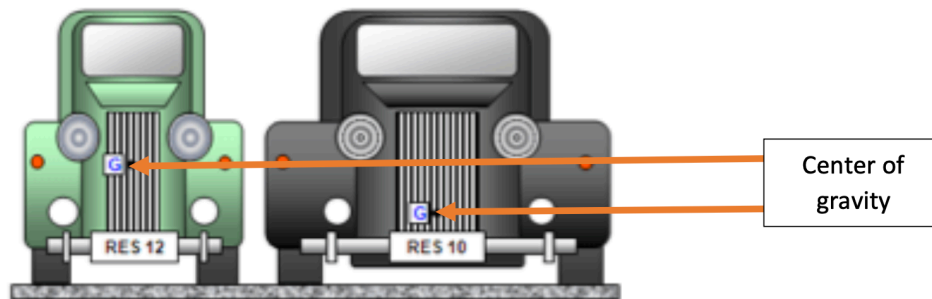
Bot B is more stable because its CG is lower (weight is mounted lower).

Bot A is less stable because its high CG means it tips more easily.

Correct statements: (Bot B more stable, CG of A is higher) and (Bot B CG is lower).

Worked Example

Two cars shown side by side. Which is unstable and why?



Answer

The left car is unstable: high centre of mass, narrow base, tall height.

Hill-Climbing Robots

To prevent a hill-climbing robot from flipping, you should put more weight at the (front/back/left/right) of the robot. This keeps the CG over the base.

Answer

Answer: Put more weight at the **front** of the robot.

This shifts the CG forward, keeping it within the base footprint on an incline.

Key Vocabulary

Term	Meaning
Centre of Gravity (CG)	The point where the object's entire weight acts
Stable	Resists tipping; CG stays over the base when tilted
Unstable	Tips easily; CG moves outside the base when tilted
Base of support	The area enclosed by the contact points (e.g. wheels)

Topic 3 — Line Following Logic

A line-following robot uses light/colour sensors to detect a black line on a white surface and stay on it. Questions test whether you understand the logic table that maps sensor readings to robot actions, and whether you can spot bugs in line-following programs.

How Light Sensors Work

- White surfaces reflect MORE light → sensor reads HIGH (white).
- Black surfaces reflect LESS light → sensor reads LOW (black).
- The robot uses these readings to know where the line is relative to its body.

The Standard 2-Sensor Logic Table

The robot has a LEFT sensor and a RIGHT sensor, positioned either side of the line. The correct logic table is:

Left Sensor	Right Sensor	Robot Action
White	White	Move Forward (both on white = centred on line)
Black	White	Turn Left (line has drifted to the left sensor)
White	Black	Turn Right (line has drifted to the right sensor)
Black	Black	Stop (reached T-junction or end of line)

How to remember this

When only ONE sensor is on black, the robot has drifted and must correct:

Left sensor on black → turn LEFT to bring the line back to centre.

Right sensor on black → turn RIGHT to bring the line back to centre.

Both white = on track → move forward.

Both black = at a junction or end → stop.

Optimal Sensor Configuration

- For curve following, you need MORE sensors spread WIDER apart.
- More sensors detect smaller deviations from the line sooner, allowing tighter corrections on curves.

Common Bugs in Line-Following Programs

Bug Type	What goes wrong
Wrong sensor used	Program reads the touch sensor instead of the colour sensor — robot never reacts to the line
Colour not checked	Loop checks sensor but never compares to 'black' — always true or always false
Condition reads outside loop	Sensor is only read once before the loop, so the value never updates
Motors at unequal speeds	Robot drifts in a curve instead of going straight
Missing brackets	while loop body not enclosed — only first line repeats

Worked Example — Spot the Bug

A program is supposed to follow a line and stop at a T-junction. The pseudocode is:

Pseudocode

```
while (read_color_sensor(2) == 'BLACK') {  
    if (read_color_sensor(1) == 'BLACK') {  
        left_motor_speed(100);  
    } else {  
        right_motor_speed(100);  
    }  
    left_motor_speed(0);  
    right_motor_speed(0);  
}
```

Bug explained

The motor stop commands are inside the while loop body but outside the if/else.

After every single pass, the motors are stopped; the robot cannot move.

Answer: The program will track the line somewhat but will not stop at the T-junction.

(Actually the robot barely moves at all due to the misplaced stop commands.)

Worked Example — Algorithm for Line Following

Which algorithm correctly describes how a car follows a line with two photoelectric sensors?

Correct algorithm

```
if the car is right above the line: go straight  
else if the car is to the RIGHT:    turn a little bit to the LEFT  
else if the car is to the LEFT:    turn a little bit to the RIGHT  
else:                               stop
```

Key: if drifted RIGHT, steer LEFT to correct. If drifted LEFT, steer RIGHT.

Topic 4 — Flowcharts

Flowcharts use standard shapes to represent different types of instructions. HCIC questions ask you to read a flowchart and identify what behaviour it describes, or match a code snippet to its flowchart.

Standard Flowchart Shapes

Shape	Name	Used for
Rounded rectangle / Oval	Start / End (Terminator)	Beginning or end of the program
Rectangle	Process	An action or calculation (e.g. Move Forward, set motor speed)
Diamond	Decision	A yes/no question — has two exit paths (Yes and No)
Parallelogram	Input / Output	Reading a sensor or printing a value
Arrow	Flow line	Shows the direction of execution

Reading a Flowchart — What to Look For

- Find the START and END — what does the program begin and end with?
- Identify all DECISION diamonds — these are the conditions (if/while).
- Trace the YES and NO paths — which action follows each outcome?
- Look for LOOPS — if a flow line goes backwards (upward) to an earlier step, it is a loop.
- Count how many times a repeated action runs — this tells you the loop count.

Worked Example — Square Route

A flowchart shows: Move Forward → 90° Left Turn → Move Forward → 90° Left Turn (repeated twice in two columns, joined at the bottom). What does this achieve?

Answer

Four repetitions of (Move Forward + 90° Left Turn) = one complete square.

Each left turn is 90°, so four turns = 360° = back to start.

Answer: The automatic car moves in a square route.

Common Flowchart Question Types on HCIC

Question type	What to do
What does this flowchart do?	Trace it from START, read each action and decision, identify the overall goal
Which flowchart matches this code?	Convert the code logic to a sequence; find the flowchart with the same sequence of shapes
How many iterations does the loop make?	Count how many times the flow line returns to the top before the exit condition is met
Is this an if or a while?	If flow goes back up: while loop. If flow only goes down: if/else.

Quick Revision Summary

Gear RPM — remember this formula

Driver Teeth × Driver RPM = Driven Teeth × Driven RPM

Smaller driven gear → faster. Larger driven gear → slower.

For a chain: solve one pair at a time.

A loop of gears (A→B→C→A) = locked, cannot move.

Stability — remember these two rules

1. Lower CG = more stable. 2. Wider base = more stable.

Heavy components high up = bad (unstable).

For hill-climbing: put weight at the FRONT.

Line Following — remember the logic table

Both white → Move Forward (on track)

Left black → Turn Left (drift left, correct left)

Right black → Turn Right (drift right, correct right)

Both black → Stop (junction or end)

Flowcharts — remember the shapes and loop rule

Diamond = decision (yes/no). Rectangle = action. Oval = start/end.

Arrow looping back upward = while loop.

Trace from START, follow YES/NO arrows, count loop iterations.

Practice Questions

Gear RPM

Q1. Gear A has 40 teeth and spins at 3 RPM. Gear B has 8 teeth and meshes with Gear A. What is Gear B's RPM? Which gear spins faster?

Q2. A 3-gear chain: Gear X (20T, 100 RPM) → Gear Y (50T) → Gear Z (10T). Find the RPM of Gear Y and Gear Z.

Q3. A robot needs to haul a heavy load up a ramp. Should the driven gear be larger or smaller than the driver gear? Explain why.

Stability

Q4. Robot A has its battery mounted at the top. Robot B has an identical battery mounted at the base. Both have the same wheel span. Which robot is more stable and why?

Q5. A robot is designed to climb steep hills. Where should the heaviest component be placed to reduce the chance of flipping? Explain using the concept of CG.

Line Following

Q6. A 2-sensor robot reads: Left = White, Right = Black. What should the robot do? Why?

Q7. A line-following program uses 'while touch_sensor is not pressed' as its loop condition. The robot is supposed to stop when it reaches a black line. What is wrong with this program?

Q8. For sharp curves, should you use 2 sensors close together or 4 sensors spread wide? Explain.

Flowcharts

Q9. A flowchart shows: Start → Move Forward → [Is obstacle within 20cm?] → Yes: Move Backward → loop back to top. No: loop back to Move Forward. What behaviour does this describe?

Q10. In a flowchart, what is the difference between a process box (rectangle) and a decision box (diamond)?

Practice Question Answers

Gear RPM Answers

Q1. $40 \times 3 = 8 \times B_RPM \rightarrow B_RPM = 120/8 = 15$ RPM. Gear B (smaller) spins faster.

Q2. $X \rightarrow Y: 20 \times 100 = 50 \times Y_RPM \rightarrow Y = 40$ RPM. $Y \rightarrow Z: 50 \times 40 = 10 \times Z_RPM \rightarrow Z = 200$ RPM.

Q3. Larger driven gear. A larger gear reduces speed but increases torque (turning force), which is needed to haul loads up inclines.

Stability Answers

Q4. Robot B is more stable. Its CG is lower because the heavy battery is at the base. Robot A's high-mounted battery raises the CG, making it easier to tip.

Q5. Place heavy components at the front (lower and forward). This keeps the CG within the base footprint even when the robot tilts on the incline, reducing the risk of flipping backwards.

Line Following Answers

Q6. Turn Right. The right sensor is on black, meaning the line has drifted to the right side. The robot must steer right to centre itself on the line.

Q7. The wrong sensor is used. The program checks the touch sensor (which requires physical contact) instead of the colour sensor (which detects black on the ground). The robot will never detect the black line.

Q8. 4 sensors spread wide. More sensors detect smaller deviations earlier, and wider spacing picks up curves before the robot has drifted too far off the line.

Flowchart Answers

Q9. Autonomous obstacle avoidance: the robot moves forward, and if an obstacle is detected within 20cm it moves backward, then tries again. It loops continuously.

Q10. A process box (rectangle) represents an action to be carried out (e.g. 'Move Forward'). A decision box (diamond) asks a yes/no question and has two exit paths — one for Yes, one for No.

End of Tier 1 Non-Python Study Guide